

Replicating Table 1 (Panels A & B) from Akhtari, Bau, and Laliberté (2020)

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1 The Original Paper

Akhtari, Mitra, Natalie Bau, and Jean-William P. Laliberté. 2020. “Affirmative Action and Pre-College Human Capital.” *NBER Working Paper* No. 27779. <http://www.nber.org/papers/w27779>

The paper studies the effect of the 2003 Supreme Court ruling in *Grutter v. Bollinger*, which reinstated race-conscious admissions at public universities in Texas, Louisiana, and Mississippi after a seven-year ban under *Hopwood v. Texas* (1996). The central research question is whether affirmative action (AA) policies affect students’ *pre-college* human capital investment—specifically SAT scores, grades, and attendance—not just outcomes *after* college admission.

The empirical strategy exploits the quasi-experimental policy reversal to estimate the average effects of AA reinstatement on SAT scores using two complementary approaches: difference-in-differences (DiD), comparing score changes in treated states (TX, LA, MS) to control states before and after 2003, and a synthetic-control method to account for differential pre-trends.

2 The Result of Interest

I replicate **Table 1, Panels A and B** (columns 1 and 2), which reports difference-in-differences estimates of the effect of AA reinstatement on mean math and verbal SAT scores—separately for underrepresented minorities (URMs: Black and Hispanic students) and for white students, using a panel of state–race–year cells from 1998 to 2010.

Panel A is the headline result: reinstating AA increased mean math SAT scores for URMs by approximately 8 points (about 0.07 standard deviations), a statistically significant

improvement. Panel B shows that white students’ math scores also increased (by about 4 points), consistent with positive spillovers or intensified competition. Both findings are among the most widely discussed results of the paper and directly test a key prediction of the theoretical model.

3 Data and Methods

Data. The analysis uses a panel of mean math and verbal SAT scores at the state–race–year level, compiled from College Board publicly available reports covering 1998–2010. The data are available via the replication package at <https://www.openicpsr.org/openicpsr/project/146381/version/V1/view>. The raw data file is `SAT_workfingfile.dta`; `code/preprocess.py` coerces stored-as-object numeric columns and writes a clean CSV to `temp/clean_data.csv`.

Identification. The identifying assumption is parallel trends: absent the policy change, URM and white SAT scores in treated states would have evolved at the same rate as in control states. The empirical model is

$$y_{ket} = \beta_{DD}(\text{TreatedState}_k \times \text{Post2003}_t) + \gamma \text{Ban}_{kt} + \alpha_k + \alpha_t + \alpha_e + \varepsilon_{ket}, \quad (1)$$

where y_{ket} is the mean SAT score for racial group e in state k in year t ; TreatedState_k is 1 for TX, LA, and MS; Post2003_t is 1 for years after 2003; Ban_{kt} controls for AA-ban policy changes in control states; and α_k , α_t , α_e are state, year, and race fixed effects. Cells are weighted by the number of test-takers, and standard errors are clustered at the state level. Equation (1) is estimated separately for URMs (Panel A) and whites (Panel B) using `code/analysis.py`, which writes the table fragment to `output/tables/table1_panels_ab.tex`.

4 Replication Results

Table 1 presents my replication.

The replication matches the paper’s Table 1 exactly across all entries. The Panel A math DD coefficient is 8.009^{***} (s.e. 1.544), and the Panel B math DD coefficient is 4.048^{***} (s.e. 0.984). The R^2 values and observation counts are identical. The analysis uses `pyfixest.feols()`, a Python port of Stata’s `reghdfe`, which applies the same degrees-of-freedom correction when computing cluster-robust standard errors. As a result, both coefficients *and* standard errors match the paper to three decimal places.

Table 1: Replication of Table 1, Panels A & B: Effect of AA on SAT Scores

	Math (1)	Verbal (2)
<i>Panel A: URMs (Black and Hispanic)</i>		
DD coefficient	8.009*** (1.544)	-0.634 (1.784)
Observations	1,904	1,901
R^2	0.844	0.796
State, Year, Race FE	X	X
<i>Panel B: Whites</i>		
DD coefficient	4.048*** (0.984)	0.034 (0.888)
Observations	663	663
R^2	0.969	0.971
State, Year, Race FE	X	X

Notes: Difference-in-differences estimates of the effect of reinstating affirmative action (AA) on mean SAT scores at the state–race–year level. The DD coefficient is the interaction of an indicator for a treated state (Texas, Louisiana, Mississippi) and an indicator for post-2003 (post-*Grutter v. Bollinger*). Regressions include controls for AA-ban policy changes in control states. Cells are weighted by the number of test-takers. Standard errors (in parentheses) are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5 What I Learned

The easiest part of the replication was matching the point estimates: once I identified the correct sample restrictions (`BH==1` or `W==1`), the correct weight variable (`takesattotalN`), and the correct control (`ban`), the DiD coefficients matched the paper exactly. The working file already contained all derived variables (treatment indicators, fixed-effects IDs, weights), so no complicated data construction was necessary.

The hardest part was data preparation. The raw `.dta` file stores numeric score and count columns as Stata string/object types because some cells contain suppressed values. Converting these to floating-point with `pd.to_numeric(errors='coerce')` was essential before any regression could run. This kind of silent type issue is easy to miss and would produce either an error or silently wrong results.

The most interesting thing I learned from doing the replication rather than just reading the paper is how important the “ban” control variable is: several control states enacted their own AA bans during the study period, and without controlling for those changes the DiD estimate conflates the *Grutter* reinstatement with those independent policy changes. The replication makes the design of the estimator far more concrete than reading about it in the

abstract.

References

- [1] Akhtari, Mitra, Natalie Bau, and Jean-William P. Laliberté. 2020. “Affirmative Action and Pre-College Human Capital.” *NBER Working Paper* No. 27779.